Lecture Notes

Lecture 5

Eigenvalues and Eigenvectors

Lecture 4 ended with an important example: in a well-chosen basis the matrix of a reflection became diagonal, $\operatorname{diag}(1,-1)$. The diagonal entries were the factors by which the operator multiplied two special directions. In this lecture we give these objects names. Directions that an operator only multiplies by a number are called *eigenvectors*, and the factors themselves are called *eigenvalues*. We will learn to find them with the characteristic polynomial. We will prove that over $\mathbb{C}$ an eigenvalue always exists. Finally, we will show that over $\mathbb{C}$ there is always a basis in which the matrix of the operator is upper triangular, with the eigenvalues on its diagonal.

Unless an example explicitly says otherwise, V is a nonzero finite-dimensional vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $T \in \mathcal{L}(V)$.

In quantum mechanics eigenvalues have a direct physical meaning. The equation $\hat{H}\psi = E\psi$ describes a state ψ whose energy is definite and equal to E. So the energy levels are the eigenvalues of the operator $\hat{H}$. In infinite-dimensional spaces an operator can also have a continuous spectrum, and the lines of an atomic spectrum show not the energies themselves but their differences. We discuss these qualifications in Lectures 10 and 14.

Learning objectives

By the end of this lecture you will be able to:

- Recognize invariant subspaces, eigenvalues, eigenvectors, and eigenspaces.
- Find eigenvalues as the roots of the characteristic polynomial.
- Find eigenspaces by Gaussian elimination.
- Check eigenvalues against the trace and the determinant.
- Explain why every operator on a nonzero finite-dimensional complex space has an eigenvalue, while over $\mathbb{R}$ it may have none.
- Prove that eigenvectors for distinct eigenvalues are linearly independent.
- Find a basis in which the matrix of a complex operator is upper triangular.

1 Invariant subspaces and eigenvectors

Guiding question. Which parts of a vector space stay unchanged when an operator acts on them? Given $T \in \mathcal{L}(V)$, we first look for the subspaces that T preserves.

Definition 5.1 (Invariant subspace). A subspace $U \subseteq V$ is called *invariant* under the operator T if T maps every vector of U to a vector of the same subspace:

$$u \in U \quad \Longrightarrow \quad Tu \in U.$$

Equivalently,

$$T(U) \subseteq U.$$

Familiar invariant subspaces. The following subspaces are always invariant under T:

$$V, \qquad \{0\}, \qquad \ker T, \qquad \operatorname{im} T.$$

The smallest nontrivial invariant subspaces are lines. For $v \neq 0$, the line $\operatorname{span}(v)$ is invariant exactly when the vector Tv is a multiple of the vector v. We call such a line an *eigenline*.

Definition 5.2 (Eigenvalue, eigenvector, eigenspace). A nonzero vector stays on its line exactly when

$$Tv = \lambda v, \qquad v \neq 0.$$

- **Eigenvalue.** A scalar $\lambda \in \mathbb{F}$ for which this equation holds is called an *eigenvalue* of T. It is the factor by which T multiplies the vector.
- **Eigenvector.** A nonzero vector v satisfying this equation is called an *eigenvector* corresponding to the eigenvalue λ. It is a direction that T preserves. An eigenvalue together with a corresponding eigenvector is called an *eigenpair*.
- **Eigenspace.** All vectors with the same eigenvalue, together with the zero vector, form the *eigenspace*

$$E(\lambda, T) = \ker(T - \lambda I) = \{v \in V : Tv = \lambda v\}.$$

The set of eigenvalues is called the *point spectrum*. The *spectrum* $\operatorname{spec}(T)$ of an operator is the set of scalars λ for which the operator $T - \lambda I$ is not invertible. In a finite-dimensional space these two sets coincide (see (5.1) below). In an infinite-dimensional space this equality can fail; we discuss this in Lecture 14.

Three equivalent statements. The eigenspace is the kernel of the operator $T - \lambda I$, so for every scalar λ the following statements are equivalent:

$$\begin{aligned} \lambda \in \operatorname{spec}(T) &\iff E(\lambda, T) \neq \{0\} \\ &\iff T - \lambda I \text{ is not injective} \\ &\iff T - \lambda I \text{ is not invertible.} \end{aligned} \tag{5.1}$$

The second equivalence is the theorem "Injectivity is a kernel condition" of Lecture 3. The third rests on the corollary "Operators: injective = surjective" of Lecture 3: on a finite-dimensional space an operator is injective if and only if it is invertible. Geometrically, an eigenvector is a direction that T only multiplies by a number (Figure 1).

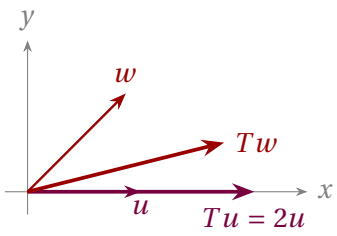


Figure 1: The eigendirection of the transformation $T(x, y) = (2x, \frac{1}{2}y)$ is the x axis: $Tu = 2u$. A general vector w is turned, so it is not an eigenvector.

Example 5.3 (Examples of eigenvalues).

- **Projection.** The eigenvalues of a projection, i.e. an operator satisfying $P^2 = P$, lie in the set $\{0, 1\}$, and its eigenspaces are $E(1, P) = \operatorname{im} P$ and $E(0, P) = \ker P$ (Figure 2).
- **Reflection.** A reflection of the plane $\mathbb{R}^2$ across a line has the eigenvalues $+1$ (for vectors on the line) and -1 (for vectors perpendicular to it), as we saw in Lecture 4 (Figure 2).
- **Differentiation.** On the space of smooth functions (which is infinite-dimensional), every scalar λ is an eigenvalue of the differentiation operator D, because $De^{\lambda t} = \lambda e^{\lambda t}$.

- **Rotation.** A rotation R_θ of the plane $\mathbb{R}^2$ with $\theta \notin \pi\mathbb{Z}$ has *no* real eigenvalues: it turns every nonzero vector (Figure 2). Over $\mathbb{C}$ the eigenvalues of the rotation are $e^{\pm i\theta}$. ◁

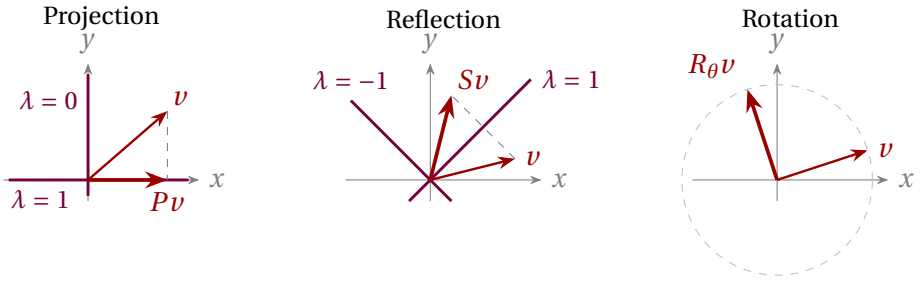


Figure 2: The projection P onto the x axis, the reflection S across the line $y = x$, and the rotation R_θ. The highlighted lines are the eigendirections, labelled with their eigenvalues. The vector v is not an eigenvector, so its image is not parallel to it. When $\theta \notin \pi\mathbb{Z}$, a rotation moves every nonzero real vector off its line, so there are no real eigenvectors.

Example 5.4 (Computing the eigenvalues of a 2×2 matrix). In the standard basis, let an operator on $\mathbb{R}^2$ have the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Let us find all its eigenvalues and eigenspaces. Before reading the solution, try to guess the eigendirections from the symmetry of the matrix.

1. **Find the eigenvalues.** By (5.1), a number λ is an eigenvalue if and only if the matrix $A - \lambda I$ is not invertible, i.e. when its determinant is zero:

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = (3 - \lambda)(1 - \lambda) = 0.$$

   So $\lambda = 3$ or $\lambda = 1$.

2. **Find the eigenspace for $\lambda = 3$.** The eigenspace $E(3, A) = \ker(A - 3I)$ is the solution set of the homogeneous system $(A - 3I)x = 0$. We reduce the matrix $A - 3I$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & -1 \\ 0 & 0 \end{bmatrix} \implies x_1 = x_2.$$

   So the free variable is x_2. Setting $x_2 = 1$, we read off the basis vector $(1, 1)$ and

$$E(3, A) = \operatorname{span}\big((1, 1)\big).$$

3. **Find the eigenspace for $\lambda = 1$.** The eigenspace $E(1, A) = \ker(A - I)$ is the solution set of the homogeneous system $(A - I)x = 0$. We reduce the matrix $A - I$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 1 \\ 0 & 0 \end{bmatrix} \implies x_1 = -x_2.$$

   So the free variable is x_2. Setting $x_2 = 1$, we read off the basis vector $(-1, 1)$. From here on it is more convenient to use its negative $(1, -1)$, which spans the same subspace:

$$E(1, A) = \operatorname{span}\big((1, -1)\big).$$

Interpretation. The two eigendirections lie along the line $y = x$ and perpendicular to it. This is what the symmetric form of the matrix A lets us guess. ◁

Example 5.5 (An invariant subspace that is not an eigenline). An invariant subspace need not be the span of an eigenvector. Consider the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- **Eigenline.** The x coordinate axis is invariant, and its nonzero vectors are eigenvectors with eigenvalue 1:
$$A(1,0) = (1,0).$$
- **Larger subspace.** The whole plane $\mathbb{R}^2$ is invariant, but it is not a line.
- **Counterexample.** The line $\operatorname{span}\big((1,1)\big)$ is not invariant, because
$$A(1,1) = (2,1) \notin \operatorname{span}\big((1,1)\big).$$

So only some lines are invariant, and these are exactly the eigenlines. ◁

2 Computing eigenvalues and eigenspaces

Guiding question. How do we find eigenvalues? In coordinates, an eigenvalue shows up as a non-invertible matrix, and invertibility can be tested with the determinant.

Definition 5.6 (Characteristic polynomial). The *characteristic polynomial* of a square matrix $A \in M_{n\times n}(\mathbb{F})$ is the polynomial

$$\chi_A(\lambda) = \det(\lambda I - A).$$

The characteristic polynomial of an operator $T \in \mathcal{L}(V)$ is the characteristic polynomial of its matrix in any basis; we show below that it does not depend on the basis. The characteristic polynomial has degree n and leading coefficient 1. Its roots are exactly the eigenvalues of A:

$$\chi_A(\lambda) = 0 \quad \Longleftrightarrow \quad \lambda \text{ is an eigenvalue of } A.$$

This follows from (5.1): the matrix $\lambda I - A$ is not invertible if and only if its determinant is zero.

Why $\lambda I - A$ and not $A - \lambda I$. In Example 5.4 we solved the equation $\det(A - \lambda I) = 0$. The matrices $\lambda I - A$ and $A - \lambda I$ differ only in sign, i.e. each of the n rows is multiplied by -1, so

$$\det(\lambda I - A) = (-1)^n \det(A - \lambda I).$$

The two determinants have the same roots, and in the definition we choose $\lambda I - A$ so that the leading coefficient is 1. The two matrices also have the same kernel, so we keep computing eigenspaces as $\ker(A - \lambda I)$.

Remark 5.7 (Determinant properties we will need). We will use four properties of the determinant, which we state without proof.

- **Invertibility.** A square matrix is invertible if and only if $\det A \neq 0$.
- **Product.** $\det(AB) = \det A \, \det B$. Hence
$$\det(P^{-1}AP) = \det A.$$

- **Triangular matrix.** The determinant of a triangular matrix is the product of its diagonal entries.
- **Coefficients.**

$$\det(\lambda I - A) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A.$$

All of them follow from expanding the determinant along a row or a column, or from row operations. The proofs can be found in the course literature.

Why the characteristic polynomial does not depend on the basis. The matrices of the same operator in different bases are related by a similarity transformation (Lecture 4). Therefore

$$\det(\lambda I - P^{-1}AP) = \det\big(P^{-1}(\lambda I - A)P\big) = \det(\lambda I - A),$$

so

$$\chi_{P^{-1}AP}(\lambda) = \chi_A(\lambda).$$

Hence the characteristic polynomial and its roots, the eigenvalues, do not depend on the chosen basis.

Proposition 5.8 (Trace and determinant from eigenvalues). Let A be a complex square matrix, and let $\lambda_1, \ldots, \lambda_n$ be its eigenvalues, i.e. the roots of the polynomial χ_A, each listed as many times as its multiplicity. Then

$$\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n, \qquad \det A = \lambda_1 \cdots \lambda_n.$$

Proof. Over $\mathbb{C}$ a polynomial factors into linear factors. This follows from the fundamental theorem of algebra, which we discuss in the proof of Theorem 5.16: having found a root, we split off a linear factor and apply the theorem to the remaining polynomial. So the same polynomial can be written in two ways:

$$\chi_A(\lambda) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A,$$

and

$$\chi_A(\lambda) = \prod_{k=1}^{n}(\lambda - \lambda_k) = \lambda^n - \left(\sum_{k=1}^{n}\lambda_k\right)\lambda^{n-1} + \cdots + (-1)^n \prod_{k=1}^{n}\lambda_k.$$

Comparing the coefficients of λ^{n-1} and the constant terms gives both identities. ∎

The method for finding eigenpairs. For a concrete matrix A we find the eigenvalues and the eigenvectors separately.

1. Form the characteristic polynomial, $\chi_A(\lambda) = \det(\lambda I - A)$.
2. Find the eigenvalues by solving the equation $\chi_A(\lambda) = 0$.
3. Check the result: compare the sum and the product of the eigenvalues with $\operatorname{tr} A$ and $\det A$ (Proposition 5.8).
4. For each eigenvalue, find the eigenspace $E(\lambda, A) = \ker(A - \lambda I)$ and a basis of it.

Example 5.9 (The 2×2 characteristic polynomial in general). Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(\mathbb{C}).$$

Then

$$\chi_A(\lambda) = \det\begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix} = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A,$$

so the eigenvalues are

$$\lambda_{\pm} = \frac{1}{2}\left(\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4\det A}\right).$$

For the matrix of Example 5.4 we get $\operatorname{tr} A = 4$ and $\det A = 3$, so

$$\chi_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The roots are the same eigenvalues 1 and 3, and their sum and product are $1 + 3 = \operatorname{tr} A$ and $1 \cdot 3 = \det A$, as Proposition 5.8 states. For a real matrix, the discriminant

$$\Delta = (\operatorname{tr} A)^2 - 4\det A$$

has a geometric meaning. When $\Delta > 0$, the matrix has two distinct real eigenvalues and two eigendirections. When $\Delta = 0$, it has one repeated real eigenvalue. When $\Delta < 0$, the eigenvalues are a pair of nonreal complex conjugate numbers. The discriminant of a complex matrix is a complex number, so there is no such classification by sign. ◁

Example 5.10 (A 3×3 characteristic polynomial). Let

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

be a lower triangular matrix. Its characteristic polynomial is

$$\chi_A(\lambda) = \det\begin{pmatrix} \lambda - 2 & 0 & 0 \\ -1 & \lambda - 3 & 0 \\ 0 & -1 & \lambda - 2 \end{pmatrix} = (\lambda - 2)(\lambda - 3)(\lambda - 2) = (\lambda - 2)^2(\lambda - 3),$$

so the eigenvalues are

$$\lambda = 2 \text{ with multiplicity } 2 \quad \text{and} \quad \lambda = 3 \text{ with multiplicity } 1.$$

The matrix $\lambda I - A$ is also triangular, so the eigenvalues can be read directly off the diagonal of A. ◁

Example 5.11 (From eigenvalues to eigenspaces). For the same matrix $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ let us find the eigenspaces $E(3, A)$ and $E(2, A)$. We compare the dimension of each with the multiplicity of the corresponding root in Example 5.10.

1. For $\lambda = 3$, the subspace $E(3, A) = \ker(A - 3I)$ is the solution set of the homogeneous system $(A - 3I)x = 0$. We reduce the matrix $A - 3I$:

$$\begin{bmatrix} -1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 \end{bmatrix} \Longrightarrow x_1 = 0, \quad x_2 = x_3.$$

   So the free variable is x_3. Setting $x_3 = 1$, we read off the basis vector $(0, 1, 1)$ and

$$E(3, A) = \operatorname{span}\big((0, 1, 1)\big).$$

2. For $\lambda = 2$, the subspace $E(2, A) = \ker(A - 2I)$ is the solution set of the homogeneous system $(A - 2I)x = 0$. We reduce the matrix $A - 2I$:

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 \end{bmatrix} \Longrightarrow x_1 = 0, \quad x_2 = 0.$$

   So the free variable is x_3. Setting $x_3 = 1$, we read off the basis vector $(0, 0, 1)$ and

$$E(2, A) = \operatorname{span}\big((0, 0, 1)\big).$$

Interpretation. The eigenvalue 2 is a double root of χ_A, but its eigenspace is only one-dimensional. Soon we will call these two numbers the *algebraic* and the *geometric multiplicity*. An operator with an eigenspace whose dimension is smaller than the multiplicity of its eigenvalue is called *defective*. Such an operator does not have enough linearly independent eigenvectors to form a basis, so its matrix is not diagonal in any basis. ◁

Example 5.12 (Complex eigenvalues of a rotation). The rotation matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

has the characteristic polynomial

$$\chi(\lambda) = (\lambda - \cos\theta)^2 + \sin^2\theta = \lambda^2 - 2\cos\theta\,\lambda + 1.$$

Its roots are

$$\lambda_\pm = \cos\theta \pm i\sin\theta = e^{\pm i\theta}.$$

When $\theta \notin \pi\mathbb{Z}$, there are no real roots, as Figure 2 shows. Over $\mathbb{C}$ both eigenvalues lie on the unit circle. ◁

3 Eigenvectors for distinct eigenvalues are independent

Guiding question. When are several eigenvectors linearly independent, so that they can become part of a basis? It is enough for their eigenvalues to be distinct.

Theorem 5.13 (Independence of eigenvectors). Suppose

$$Tv_j = \lambda_j v_j \qquad (j = 1, \dots, m),$$

where each v_j is nonzero and the eigenvalues $\lambda_1, \dots, \lambda_m$ are distinct. Then

$$v_1, \dots, v_m \quad \text{are linearly independent.}$$

Proof. We prove the theorem by contradiction. Suppose the vectors $v_1, \dots, v_m$ are linearly dependent. We take the first vector that is a linear combination of the preceding ones and use the eigenvalue equation to show that this cannot happen.

1. By the linear dependence lemma of Lecture 2, there is a smallest index k such that

   $$v_k \in \operatorname{span}(v_1, \dots, v_{k-1}).$$

   Then the vectors $v_1, \dots, v_{k-1}$ are linearly independent, and there are scalars a_j such that

   $$v_k = \sum_{j<k} a_j v_j.$$

2. We apply the operator T and use $Tv_j = \lambda_j v_j$:

   $$\lambda_k v_k = \sum_{j<k} a_j \lambda_j v_j.$$

3. We multiply the original equation by λ_k:

   $$\lambda_k v_k = \sum_{j<k} a_j \lambda_k v_j.$$

4. We subtract one equation from the other:

$$0 = \sum_{j<k} a_j(\lambda_j - \lambda_k)v_j.$$

5. The vectors $v_1, \ldots, v_{k-1}$ are linearly independent, so

$$a_j(\lambda_j - \lambda_k) = 0 \qquad (j < k).$$

The eigenvalues are distinct, so every factor $\lambda_j - \lambda_k$ is nonzero, and hence all $a_j = 0$. But then $v_k = 0$, which contradicts the definition of an eigenvector. ∎

What this theorem gives. Eigenvectors for distinct eigenvalues are always linearly independent. There is no need to check their independence by Gaussian elimination.

Corollary 5.14 (Number of eigenvalues). An operator on an n-dimensional space has at most n distinct eigenvalues.

Proof. Eigenvectors for distinct eigenvalues form a linearly independent list. By the theorem "Independent ≤ spanning" of Lecture 2, the length of such a list is at most $n = \dim V$. ∎

Corollary 5.15 (The sum of eigenspaces is direct). If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, then

$$E(\lambda_1, T) + \cdots + E(\lambda_m, T) = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Proof. Suppose we have written the zero vector as a sum

$$0 = u_1 + \cdots + u_m, \quad \text{where } u_j \in E(\lambda_j, T) \text{ and not all } u_j = 0.$$

The nonzero terms are eigenvectors for distinct eigenvalues, and their sum is zero. This contradicts Theorem 5.13. So the zero vector can be written only trivially, and the sum is direct. ∎

Connection with diagonalization. Theorem 5.13 gives the following chain of implications when $\dim V = n$:

$$\begin{aligned} n \text{ distinct eigenvalues} &\implies n \text{ independent eigenvectors} \\ &\implies \text{a basis of eigenvectors.} \end{aligned}$$

In a basis of eigenvectors the matrix of the operator is diagonal. We call an operator *diagonalizable* if it has a basis of eigenvectors. We derive a general criterion for diagonalizability in Lecture 6.

4 Existence of eigenvalues over $\mathbb{C}$

Guiding question. Why can a real rotation have no eigenvector, while a complex operator is guaranteed to have one? It all depends on whether the characteristic polynomial has at least one root.

Theorem 5.16 (Eigenvalues exist over $\mathbb{C}$). If

$$V \neq \{0\}, \qquad \dim V < \infty, \qquad V \text{ is a complex vector space,}$$

then every $T \in \mathcal{L}(V)$ has an eigenvalue.

Proof. Choose a basis of V and let A be the matrix of T in it. We rely on the *fundamental theorem of algebra*: every nonconstant polynomial with complex coefficients has a complex root. We do not prove this theorem, because its proof rests on analysis, not on linear algebra. The proof is the following chain of implications:

$$\begin{aligned}\deg \chi_A = \dim V \geq 1 &\xRightarrow{\text{fundamental theorem of algebra}} \chi_A(\lambda) = 0 \text{ for some } \lambda \in \mathbb{C} \\ &\Longrightarrow \det(\lambda I - A) = 0 \\ &\Longrightarrow \ker(\lambda I - A) \neq \{0\} \\ &\Longrightarrow Av = \lambda v \text{ for some } v \neq 0.\end{aligned}$$

So λ is an eigenvalue of T. ∎

Remark 5.17.

- **Where $\mathbb{C}$ is needed.** The fundamental theorem of algebra guarantees that a root exists. A real polynomial need not have a real root.
- **What the theorem does not say.** In quantum mechanics complex numbers describe phase and interference. But the existence of eigenvalues alone does not yet explain measurement. Which operators correspond to measurable quantities, and with what probability each outcome is obtained, we discuss in Lectures 9–10.

5 Multiplicity and defective operators

Guiding question. Does an eigenvalue repeated k times always give k linearly independent eigenvectors?

Two multiplicities. For each eigenvalue λ we define two numbers:

- The **algebraic multiplicity** $\text{mult}_{\chi_T}(\lambda)$ is the multiplicity of the eigenvalue λ as a root of the polynomial χ_T.
- The **geometric multiplicity** is the dimension of the eigenspace

$$\dim E(\lambda, T) = \dim \ker(T - \lambda I).$$

Example 5.11 shows that these numbers can differ.

Proposition 5.18 (Geometric ≤ algebraic). For every eigenvalue λ,

$$\dim E(\lambda, T) \leq \text{mult}_{\chi_T}(\lambda).$$

Proof (self-study). Let $g = \dim E(\lambda, T)$.

1. Let $v_1, \ldots, v_g$ be a basis of $E(\lambda, T)$. Extend it to a basis $v_1, \ldots, v_n$ of V.
2. Since $Tv_k = \lambda v_k$ for $k \leq g$, the matrix of T in this basis has the block form
$$\mathcal{M}(T) = \begin{pmatrix} \lambda I_g & B \\ 0 & C \end{pmatrix}.$$
3. This matrix is block upper triangular, so its determinant is the product of the determinants of the diagonal blocks. This can be seen by expanding the determinant along the first column: because of the zero block, only one term survives at each step. So the characteristic polynomial factors as
$$\chi_T(t) = \det(tI_g - \lambda I_g)\det(tI_{n-g} - C) = (t - \lambda)^g \det(tI_{n-g} - C).$$

4. Hence the polynomial $\chi_T(t)$ is divisible by $(t-\lambda)^g$. So the multiplicity of λ as a root is at least g. ∎

When diagonalization fails. An operator is called *defective* if the geometric multiplicity of at least one of its eigenvalues is smaller than the algebraic one. The standard example is the shear

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Its only eigenvalue 1 has the multiplicities

$$\underbrace{\operatorname{mult}_{\chi_S}(1)}_{2} > \underbrace{\dim E(1,S)}_{1}, \qquad E(1,S) = \operatorname{span}\big((1,0)\big).$$

So the shear does not have enough eigenvectors to form a basis.

6 Upper triangular matrices

Guiding question. What can we do when an operator does not have enough eigenvectors to form a basis? Over $\mathbb{C}$ it is still possible to find a basis in which the matrix of the operator is upper triangular.

Invariant flag. An ordered basis defines nested subspaces

$$\operatorname{span}(v_1) \subseteq \operatorname{span}(v_1, v_2) \subseteq \cdots \subseteq \operatorname{span}(v_1, \ldots, v_n) = V.$$

Such a chain of subspaces is called a *flag*. The matrix is upper triangular exactly when the operator T leaves every subspace of this flag invariant.

Proposition 5.19 (Triangular matrices and invariant subspaces *[self-study]*). Let $v_1, \ldots, v_n$ be a basis of V. The following statements are equivalent:

$$\begin{aligned} &\mathcal{M}(T, (v_1, \ldots, v_n)) \text{ is upper triangular} \\ \iff\ & Tv_k \in \operatorname{span}(v_1, \ldots, v_k) \quad (k = 1, \ldots, n) \\ \iff\ & \operatorname{span}(v_1, \ldots, v_k) \text{ is invariant for every } k. \end{aligned}$$

Proof. The k-th column of the matrix contains the coordinates of the vector Tv_k. The entries of this column below the k-th row are zero exactly when

$$Tv_k \in \operatorname{span}(v_1, \ldots, v_k).$$

This is the first equivalence. If the subspace $\operatorname{span}(v_1, \ldots, v_k)$ is invariant, then the vector Tv_k belongs to it. Conversely, if the second condition holds for all k, then for every $j \le k$ the vector Tv_j belongs to the subspace $\operatorname{span}(v_1, \ldots, v_j) \subseteq \operatorname{span}(v_1, \ldots, v_k)$. So the subspace $\operatorname{span}(v_1, \ldots, v_k)$ is invariant. ∎

Theorem 5.20 (Triangularizability over $\mathbb{C}$ ·sketched). For every operator T on a finite-

dimensional complex vector space there is a basis in which

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}.$$

Proof sketch (self-study). We use induction on $n = \dim V$.

1. By Theorem 5.16, the operator T has an eigenvector v_1. Its line $\operatorname{span}(v_1)$ is invariant.
2. Consider the $(n-1)$-dimensional quotient space $V/\operatorname{span}(v_1)$ (Lecture 3). The line $\operatorname{span}(v_1)$ is invariant, so the formula
$$\bar{T}\big(v + \operatorname{span}(v_1)\big) = Tv + \operatorname{span}(v_1)$$
gives a well-defined operator on this quotient space.
3. By the induction hypothesis, the quotient space has a basis $\bar{v}_2, \dots, \bar{v}_n$ in which the matrix of this operator is upper triangular.
4. Choose a representative of each coset; this gives vectors $v_2, \dots, v_n \in V$. Together with v_1 they form a basis of V that satisfies the condition
$$Tv_k \in \operatorname{span}(v_1, \dots, v_k).$$
5. By Proposition 5.19, the matrix is upper triangular in this basis.

Complex numbers are needed only to find an eigenvalue at each step. So the same proof works over any field in which the polynomial χ_T splits into linear factors. ∎

Eigenvalues on the diagonal. The eigenvalues of a triangular matrix can be read directly off its diagonal.

Theorem 5.21 (Eigenvalues sit on the diagonal). If T has an upper triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$\operatorname{spec}(T) = \{\lambda_1, \dots, \lambda_n\}.$$

Moreover,

$$T \text{ is invertible} \iff \lambda_k \neq 0 \text{ for every } k.$$

Proof. Let A be the upper triangular matrix of T. Then the matrix $\lambda I - A$ is also upper triangular, so by Remark 5.7

$$\chi_A(\lambda) = \det(\lambda I - A) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n).$$

The roots of this polynomial are exactly the diagonal entries. Also,

$$\det A = \lambda_1 \cdots \lambda_n,$$

and this product is nonzero exactly when every diagonal entry is nonzero. A matrix is invertible if and only if its determinant is nonzero. ∎

Example 5.22 (Reading eigenvalues off a triangular matrix). Let us read off the eigenvalues of the matrix

$$A = \begin{pmatrix} 3 & 7 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}$$

and decide whether it is invertible. There is no need to expand the determinant.

1. We read off the diagonal:
$$\lambda_1 = 3, \quad \lambda_2 = 3, \quad \lambda_3 = -1.$$
So the eigenvalue 3 has algebraic multiplicity 2, and the eigenvalue -1 has algebraic multiplicity 1.
2. We check invertibility:
$$\det A = 3 \cdot 3 \cdot (-1) = -9 \neq 0.$$
Hence the matrix A is invertible. ◁

Connection to physics: observables

In the last example we will see what eigenvalues mean in quantum mechanics.

Example 5.23 (The Pauli matrices σ_x and σ_y). The spin projections onto the x and y axes, measured in units of $\hbar/2$, are described by the Pauli matrices
$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$
Both matrices have the characteristic polynomial $\chi(\lambda) = \lambda^2 - 1$, and their eigenvalues are $+1$ and -1. The eigenpairs with normalized eigenvectors (of length 1) are
$$\sigma_x: \quad \lambda = +1,\ v_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \lambda = -1,\ v_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix},$$
$$\sigma_y: \quad \lambda = +1,\ w_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix} \quad \text{and} \quad \lambda = -1,\ w_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Physical meaning. When the spin projection onto the x axis is measured, only two outcomes are possible, $+1$ and -1. These are the eigenvalues of the matrix σ_x. The corresponding eigenvectors $v_\pm$ describe the states in which the outcome of the measurement is known in advance. The vectors $w_\pm$ describe spin states pointing along the $+y$ and $-y$ axes. The eigenvectors $w_\pm$ of the matrix σ_y also have an imaginary coordinate, so the space of quantum states is naturally complex. We discuss the probabilities of measurement outcomes in Lecture 10. ◁

Summary of main concepts

In an eigenpair, the eigenvalue is a factor and the eigenvector is a direction:
$$Tv = \lambda v, \qquad E(\lambda, T) = \ker(T - \lambda I).$$
In finite dimension, λ is an eigenvalue exactly when $T - \lambda I$ is not invertible.

The roots of the characteristic polynomial are the eigenvalues, and we find the eigenvectors by computing kernels with Gaussian elimination. Eigenvectors for distinct eigenvalues are linearly independent. If an operator on an n-dimensional space has n distinct eigenvalues, then there is a basis of eigenvectors.

Over $\mathbb{C}$ the fundamental theorem of algebra guarantees at least one eigenvalue. Repeating this argument with invariant subspaces, we get a basis in which the matrix of the operator is upper triangular, with all the eigenvalues on its diagonal.

- **Eigenvalue, eigenvector, eigenspace**—$Tv = \lambda v$; $E(\lambda, T) = \ker(T - \lambda I)$; in finite dimension, λ is an eigenvalue $\iff$ $T - \lambda I$ is not invertible.

- **Independence of eigenvectors**—eigenvectors for distinct eigenvalues are linearly independent; there are at most $\dim V$ distinct eigenvalues.
- **Eigenvalues exist over** $\mathbb{C}$—every operator on a nonzero finite-dimensional complex space has an eigenvalue (fundamental theorem of algebra).
- **Characteristic polynomial**—$\chi_A(\lambda) = \det(\lambda I - A)$; its roots are the eigenvalues, and $\operatorname{tr} A = \sum \lambda_k$, $\det A = \prod \lambda_k$.
- **Triangularizability over** $\mathbb{C}$—in a suitable basis the matrix of a complex operator is upper triangular, with the eigenvalues on its diagonal.